



Marked for the Camera: A Staff Survey on Deliberate Whiteboard Use Under a Meeting-Room Engagement Photo-Audit

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Abstract. Since its introduction, Facilities' Room Engagement Score has scored every booking across the University's centrally bookable meeting rooms from a single photograph of the room's whiteboard, taken by the booking system's own camera at the end of each slot: a mark on the board reads as "engaged," a bare board reads as "engagement not evidenced," and a room's running engaged-rate sets its priority in the following term's booking allocation. We surveyed 138 staff holding standing bookings across 46 scored rooms on whether, and why, they mark the board, and matched self-reported behaviour against the Score's own photo-audit log and an independently coded record of whether each meeting reached a decision. Self-reported deliberate marking is common in high-demand rooms (61% report marking "usually" or "always," against 19% in low-demand rooms) and strongly predicts a booking's audit-engaged status (OR = 4.8, 95% CI 2.6-8.9); it is only weakly related to whether the independently coded record shows a decision was reached ($r = 0.09$, n.s.). Comparing coded decision-density between marked and unmarked meetings directly leaves the two distributions statistically indistinguishable. We suggest the Score functions, in the rooms and term we sampled, as a room-allocation signal staff have learned to produce on demand, largely independent of what a meeting in fact decided.

Introduction

Facilities' Room Engagement Score has, since its introduction, scored a booking not by who attended or what was decided but by what the room's whiteboard looks like when the booking system's camera photographs it at the end of the slot. A classifier reads the photograph for marker-pixel coverage above a fixed threshold; a room clears the threshold or it does not, and a room's running share of cleared bookings sets its standing in the following term's allocation review, which reassigns persistently low-scoring rooms to teaching pods and quiet-study use. The Score was adopted, per Facilities' own documentation, as a low-cost proxy for whether a booked room was doing what a meeting room is for: a photographed board is the nearest thing to evidence a camera aimed at an empty room after the fact can gather.

A proxy adopted for its cheapness invites a second question the adoption decision does not itself answer: what, specifically, produces the mark. A whiteboard fills with ink when a meeting works something out on it, but it also fills with ink when someone in the room knows the photograph is coming and wants the room to keep its slot. Nothing in the Score's design distinguishes the two; nothing in Facilities' allocation review has, to our knowledge, been asked to.

We report a survey of staff holding standing bookings in Score-covered rooms, asking directly whether and why they mark the board, set against the Score's own audit log and an independently coded record of whether each surveyed meeting reached a decision. Our contributions, each hedged as a single-term staff survey warrants:

- We provide survey evidence that deliberate, booking-motivated whiteboard marking is common, and concentrated in the rooms with most to lose from a low score.
- We show self-reported deliberate marking strongly predicts a booking's audit-engaged status but is only weakly related to an independently coded record of whether the meeting reached a decision.
- We characterise the meeting types most likely to be marked defensively, and the term-cycle timing at which defensive marking is most concentrated.

Related work

Whether a measure adopted as a proxy comes to be optimised in place of the thing it was meant to track is among the oldest observations in the audit and evaluation literature: quantitative indicators used for social decision-making are, on Campbell's original account, apt to distort the processes they monitor once decisions ride on them [1], and English public-health targets have been shown to generate exactly the gaming his account predicts [2]. Higher education is not exempt from its own instruments: the audit culture Strathern traces through the British university system reshapes what a department presents for review independent of what the review was meant to assess [3], a pattern Power's later account of the "audit society" treats as general to any regime built on externally verified indicators [4], and formal work on proxy-based competition shows the degradation is a structural property of optimising against a stand-in for a genuine, costlier-to-measure target rather than a merely anecdotal one [5].

The engineering literature on meeting-room sensing is largely occupied with detection accuracy — occupancy classifiers trained against ad-hoc booking and infrared data [6], [7], and IoT deployments validating a space-use signal against a university's own booking log [8] — without asking what a booking population does once it learns a detector is watching. The activity-based-working literature comes closer to our question without quite reaching it: post-occupancy studies tie a utilisation signal to satisfaction, productivity, and physical activity [9], [10], a communication-behaviour study of a renovated research facility treats occupants' space use as a stable trait of the redesign rather than a response to being counted [11], and workspace-

preference fieldwork documents staff actively avoiding or claiming particular desks for reasons the utilisation figures alone would not surface [12] — evidence that occupants routinely act on a space differently than a headline utilisation number would suggest, without asking whether the number itself supplied the incentive.

Work on visibility and observation offers the closer analogue: increased observability has been shown to change what workers do rather than simply reveal it [13], research-participation effects persist as a live methodological concern half a century after their original description [14], and the durable gap between what a self-report captures and what direct behavioural observation would show is a standing caution against treating either alone as ground truth [15]. Survey evidence on workplace surveillance more broadly finds perceived monitoring itself, not only its consequences, shapes worker well-being and trust [16], and participatory work with sensor-tracked staff shows workers actively contesting and reinterpreting what a tracking system is taken to mean once they are shown its output [17] — both consistent with a monitored population treating the instrument, not only the activity it records, as something to manage.

Method

The Room Engagement Score covers 46 centrally bookable meeting rooms across the precinct Facilities' booking system manages directly (school-held rooms booked through local calendars are out of scope). At the end of each booked slot, a ceiling-mounted camera captures one photograph of the room's whiteboard; a classifier computes the fraction of board pixels exceeding a fixed marker-ink threshold and returns a binary label:

```
delta = marker_pixel_fraction(photo)
if delta > MARK_THRESHOLD:
    status = "engaged"
else:
    status = "engagement not evidenced"
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A room's engaged-rate over the preceding term — the share of its bookings labelled "engaged" — sets its standing in Facilities' term-end allocation review; rooms in the lowest-scoring band are flagged for reassignment to teaching pods or quiet-study use the following term.

We surveyed all 138 staff (86 academic, 52 professional) holding at least one standing

weekly or fortnightly booking in a Score-covered room during the audited term, achieving a 71% response rate. The survey asked how often respondents mark the board during a booked meeting (never/rarely/sometimes/ usually/always), whether marking is done “for the meeting” or “in case the room is checked,” and the respondent’s own room’s approximate booking-demand tier (self-reported, cross-checked against the booking system’s occupancy figures). We classify a room as high-demand if booked for more than 80% of its available slots across the term and low-demand otherwise.

We obtained the Score’s own photo-audit log for all 612 bookings made by surveyed respondents across the term, giving each booking’s classifier label directly. For a random subsample of 240 of these bookings we additionally obtained the minute-taker’s own record, where one was kept, and had two independent raters code whether the record showed the meeting reached a decision (agreed on an action, allocation, or recommendation), resolving disagreement by discussion (Cohen’s $\kappa = 0.81$).

We modelled the probability of a booking’s audit-engaged label as a logistic function of the respondent’s self-reported deliberate-marking frequency (collapsed to a binary “usually/always” versus lower) and room-demand tier:

$$\text{logit}(P(\text{engaged})) = \beta_0 + \beta_1 \cdot \text{marks} + \beta_2 \cdot \text{demand tier}$$

and separately estimated the point-biserial correlation between self-reported deliberate marking and the independently coded decision outcome, across the 240-booking decision-coded subsample.

Results

Self-reported deliberate whiteboard marking is common, and unevenly distributed. Across the 138 respondents, 61% holding a standing booking in a high-demand room report marking the board “usually” or “always,” against 19% in low-demand rooms (Figure 1). Asked directly why, respondents in high-demand rooms were markedly more likely to select “in case the room is checked” over “for the meeting” as their primary reason; the reverse held in low-demand rooms, where few respondents reported the Score as a consideration at all.

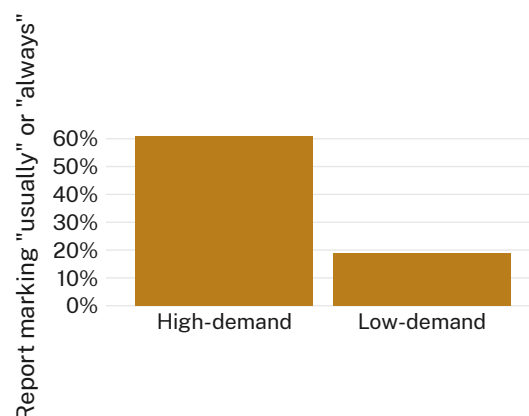


Figure 1: Self-reported deliberate whiteboard marking (“usually” or “always”), by room booking-demand tier: 61% in high-demand rooms against 19% in low-demand rooms.

The logistic model confirms the association holds at the level of individual bookings, not just self-reported habit: self-reported deliberate marking strongly predicts a booking’s audit-engaged label (OR = 4.8, 95% CI 2.6-8.9, $p < 0.01$), controlling for room-demand tier. The Score’s own engaged-rate has risen accordingly since its introduction, from 38% in the first audited term to 84% in the seventh (Figure 2); the independently coded decision-density of the same bookings — the mean number of decisions per meeting, drawn from the 240-booking coded subsample available in each term — shows no corresponding rise, holding at 2.1-2.3 decisions per meeting throughout.

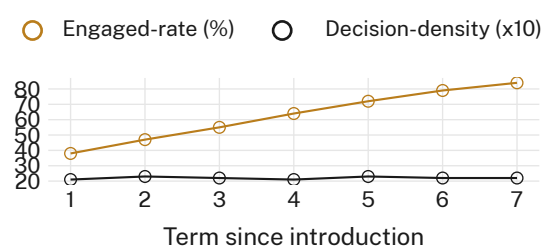


Figure 2: The Score’s engaged-rate has risen from 38% to 84% over seven terms; independently coded decision-density per meeting has not moved.

Self-reported deliberate marking is only weakly related to whether a meeting reached a decision: across the 240-booking coded subsample, the point-biserial correlation between deliberate marking and coded decision outcome is $r = 0.09$ (n.s.). Comparing coded decision-density directly between marked and unmarked bookings leaves the two distributions statistically indis-

tinguishable (medians 2.2 and 2.2 decisions; Figure 3), an ablation that, run the obvious way, changes nothing: whatever predicts a marked board, it is not, on this sample, whether the meeting decided something.

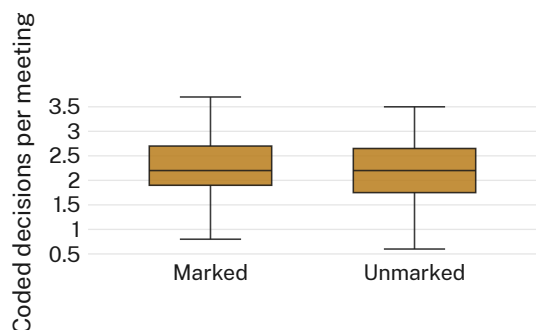


Figure 3: The ablation that changes nothing: coded decision-density for marked and unmarked bookings holds at the same median (2.2 vs 2.2).

Two further patterns round out the record. Defensive marking, by respondents' own account, concentrates in the final three weeks of term, immediately ahead of Facilities' allocation review; several respondents in high-demand rooms described marking a board at the start of a meeting specifically, "before starting," rather than during or after it. And low-demand-room respondents who did report deliberate marking cited a different motive almost without exception: not preserving their own room, but making a currently under-booked room look more used than it is, in the hope of avoiding a term where it disappears from the bookable list entirely.

Discussion

Our results suggest the Score functions, in the rooms and term we sampled, as a signal staff have learned to produce largely independent of what a meeting in fact accomplishes, most consistently so in exactly the high-demand rooms whose allocation the Score exists to protect. This need not mean the Score was wrongly designed for its narrower original purpose — confirming a room was used at all rather than sitting empty — so much as that a photograph taken because it was cheap to take has come to carry more allocation weight than a cheap photograph can reasonably bear. The timing concentration around Facilities' term-end review is consistent with a proxy staff read correctly as consequential rather than incidental, and the low-demand-room finding suggests the gaming runs in both

directions: some staff protect a room they already hold, others manufacture use in a room they are trying to keep. We observe the pattern is strongest in exactly the settings where the allocation stakes are highest, which is also where a defensive-marking account is least surprising and, on that reading, least in need of further explanation.

Limitations

Our survey covers one term and the 46 centrally bookable rooms Facilities' Score directly scores, and we make no claim the marking pattern generalises to school-held rooms outside its scope, though the consistency of the demand-tier split across both academic and professional respondents suggests the mechanism is not specific to either group. Decision-coding was available for 240 of 612 audited bookings, limited by minute-taking practice rather than by our sampling; the coded subsample's demand-tier composition matched the full sample closely enough that we do not think this materially biases the comparison. Self-report of marking motive is, inevitably, itself a self-report, though the convergence between stated motive and audit-log timing is not something a purely self-serving account would need to produce.

Conclusion

A meeting-room engagement measure built from a single end-of-slot photograph of a whiteboard is, on the evidence of one term's staff survey and audit log, more reliably predicted by whether staff have learned the photograph is coming than by whether the meeting it photographs reached a decision. Self-reported deliberate marking predicts a room's audit-engaged status strongly and its independently coded decision-density not at all. Facilities' term-end allocation review currently treats the Score's engaged-rate as a proxy for productive room use; our results suggest it is, at minimum, also a proxy for which rooms' occupants know the review is coming. Future work might extend the audit beyond a single term, or test whether relocating the camera changes what gets marked.

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